

220 Quiz 2

Question 1

To estimate the average weight of all Chihuahuas, we took a random sample with replacement of 1,000 Chihuahuas from around the world, weighed them, and calculated the average weight to be 3.9 pounds. We report a 95% confidence interval of $[3.6, 4.2]$. Which of the following is true? Select all that apply.

- (a) There is a 95% chance that the average weight of all Chihuahuas in the world is between 3.6 and 4.2 pounds.
- (b) If we repeated this procedure many times, 95% of the intervals calculated this way will include the mean weight of all Chihuahuas in the world.
- (c) The interval $[3.6, 4.2]$ corresponds to the middle 95% of the bootstrap estimated sampling distribution.
- (d) There is a 95% chance the average weight of all Chihuahuas in the world is 3.9 pounds.

Question 2

Below is a proof using properties/rules of expectations. Select the correct rule being used at each point.

Prove that $\hat{\mu} = \frac{1}{n} \sum_{i=1}^n Y_i$ is unbiased for the population mean μ : $E[\hat{\mu}] = \mu$

$$E[\hat{\mu}] = E\left[\frac{1}{n} \sum_{i=1}^n Y_i\right] \quad \text{take expectation of estimator}$$

$$\frac{1}{n} E\left[\sum_{i=1}^n Y_i\right] \quad \begin{array}{l} \text{(a) Constant Rule} \\ \text{(b) Linearity of Expectations} \\ \text{(c) Irrelevant Conditioning} \\ \text{(d) Law of Iterated Expectations} \end{array}$$

$$\frac{1}{n} E[Y_1 + Y_2 + Y_3 + \dots + Y_n]$$

$$\frac{1}{n} (E[Y_1] + E[Y_2] + E[Y_3] + \dots + E[Y_n]) \quad \begin{array}{l} \text{(a) Constant Rule} \\ \text{(b) Linearity of Expectations} \\ \text{(c) Irrelevant Conditioning} \\ \text{(d) Law of Iterated Expectations} \end{array}$$

$$\frac{1}{n} (E[Y] + E[Y] + E[Y] + \dots + E[Y]) \text{ (n times)} \quad \text{Y's are IID}$$

$$\frac{1}{n} (\mu + \mu + \mu + \dots + \mu) \text{ (n times)} \quad \text{by definition}$$

$$\frac{1}{n} \sum_{i=1}^n \mu$$

$$\frac{1}{n} n\mu$$

$$= \mu$$